

entropy

Vincent Wang

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1 1.1 Entropy

Entropy: $H(X) = -E[\log_2 P_X(X)] = -\sum_x P_X(x) \log_2(P_X(x))$. Joint entropy:

$$H(X_1, X_2, \dots, X_n) = -E[\log_2 P_{X_1, \dots, X_n}(X_1, \dots, X_n)] = -\sum_{x_i} P_{X_i}(x_i) \log_2 P_{X_i}(x_i).$$

Conditional entropy:

$$\begin{aligned} H(X|Y) &= -E[P_{X|Y}(X)] = \sum_y P_Y(y) H(X|Y=y) \\ &= -\sum_y P_Y(y) \sum_x P(X=x|Y=y) \log_2 P(X=x|Y=y) \end{aligned}$$

Bernoulli:

$$H(B_p) = -p \log_2 p - (1-p) \log_2 (1-p).$$

Consider $f(x) = x \log x \rightarrow f'(x) = 1 + \log x \rightarrow f''(x) = 1/x > 0$, so f is concave up. Suppose X takes n values $\{x_i\}_{i \in [n]}$. Then

$$\begin{aligned} \frac{1}{n} \sum_x P_X(x) \log_2 P_X(x) &\geq \frac{1}{n} \log_2 \frac{1}{n} \\ \sum_x P_X(x) \log_2 P_X(x) &\geq -\log_2 n \\ H(X) &\leq \log_2 n \end{aligned}$$

(by Jensen's).

Geometric: Let $P(X=i) = pq^i$, $i \in \mathbb{Z}_{\geq 0}$. Then

$$\begin{aligned} H(X) &= -\sum_{i=0}^{\infty} pq^i \log_2(pq^i) = -\sum_{i=0}^{\infty} pq^i \log_2 p - \sum_{i=0}^{\infty} pq^i i \log_2 q \\ &= -\log_2 p - p \log_2 q \cdot \frac{q}{p^2} = -\log_2 p - \frac{q}{p} \log_2 q = -\frac{p \log_2 p - q \log_2 q}{p} = h(p)/p. \end{aligned}$$

Sps $p = 1$. Then $H(X) = 1$ (expected). Sps $p = 1/2$. Then $H(X) = 2 \log_2 2$. Note $H(G_p) \geq H(B_p)$.

$H(X) = \infty?$. Take $P(X = k) = \frac{c}{k \log^2 k}$ for $k = 2, 3, \dots$. Note that

$$\int \frac{1}{x \log^2 x} dx = -\frac{1}{\log x},$$

so $\sum_{k=2}^{\infty} \frac{1}{k \log^2 k} = \frac{1}{c} < \infty$. Compute

$$\begin{aligned} H(X) &= -\sum_{k=2}^{\infty} \frac{c}{k \log^2 k} \log \left(\frac{c}{k \log^2 k} \right) \\ &= \sum_{k=2}^{\infty} \frac{c}{k \log^2 k} \log \left(\frac{k \log^2 k}{c} \right) \\ &\geq \sum_{k=k_0}^{\infty} \frac{c}{k \log^2 k} \log \left(\frac{k \log^2 k}{c} \right) \geq \sum_{k=k_0}^{\infty} \frac{c}{k \log^2 k} \log k \\ &= \sum_{k=k_0}^{\infty} \frac{c}{k \log k}. \end{aligned}$$

Note that

$$\int \frac{dx}{x \log x} = \log \log x,$$

so $H(x) = \infty$.

Let f be convex. Then $f(E[X]) \leq E[f(X)]$. Proof:

$$f(E[X]) = f\left(\sum_x P_X(x)x\right) \leq \sum_x P_X(x)f(x) = E[f(X)].$$

Sps f is strictly convex and E is not constant. Suppose $\text{supp } E = \{x, y\}$. Let the assoc probs be p, q (so $p, q \neq 0$). Then

$$f(E[X]) = f(px + qy) < pf(x) + qf(y) = E[f(X)].$$

$H(X) = 0$ iff $X = \{x_0\}$. Proof: If $P_X(x_0) = p \in (0, 1)$ for some $x_0 \in X$, then $-p \log p > 0$, absurd. Hence, $P_X(x_0) = 1$, so $X = \{x_0\}$.

Conditioning reduces entropy: Let X, Y be rvs. Then $H(X|Y) \leq H(X)$, with equality iff X, Y are independent.

Proof:

$$\begin{aligned}
H(X|Y) &= \sum_y P_Y(y) H(X|Y=y) = \sum_y P_Y(y) \sum_x P_{X|Y}(x,y) \log_2 \frac{1}{P_{X|Y}(x,y)} \\
&= \sum_x \sum_y P_Y(y) P_{X|Y}(x,y) \log_2 \frac{1}{P_{X|Y}(x,y)} \\
&= \sum_x \sum_y P_X(x) P_{Y|X}(x,y) \log_2 \frac{P_Y(y)}{P_{XY}(x,y)} \\
&= \sum_x P_X(x) \sum_y P_{Y|X}(x,y) \log_2 \frac{P_Y(y)}{P_{XY}(x,y)} \\
&\leq \sum_x P_X(x) \log_2 \left(\sum_y \frac{P_{Y|X}(x,y) P_Y(y)}{P_{XY}(x,y)} \right) \\
&= \sum_x P_X(x) \log_2 \frac{1}{P_X(x)} = H(X),
\end{aligned}$$

since $\log : (0, \infty) \rightarrow \mathbb{R}$ is concave down. Equality holds iff $\frac{P_Y(y)}{P_{XY}(x,y)}$ is constant for fixed x and variable y .

Specifically,

$$\frac{P_Y(y)}{P_{XY}(x,y)} = \frac{\sum_y P_Y(y)}{\sum_y P_{XY}(x,y)} = \frac{1}{P_X(x)},$$

so $P_{XY}(x,y) = P_X(x)P_Y(y)$ for all x, y , ie X, Y are independent.

Claim: for all rvs X, Y , we have $H(X, Y) \leq H(X) + H(Y)$.

Proof:

$$\begin{aligned}
H(X|Y) &= \sum_y P_Y(y) H(X|Y=y) = - \sum_y \sum_x P_Y(y) P(X=x|Y=y) \log_2 P(X=x|Y=y) \\
&= - \sum_y \sum_x P_Y(y) \frac{P_{XY}(x,y)}{P_Y(y)} \log_2 \frac{P_{XY}(x,y)}{P_Y(y)} = \sum_y \sum_x P_{XY}(x,y) \log_2 \frac{P_Y(y)}{P_{XY}(x,y)} \\
&= H(X, Y) + \sum_y P_Y(y) \log_2 P_Y(y) = H(X, Y) - H(Y).
\end{aligned}$$

Combining with the previous result, we have $H(X, Y) \leq H(X) + H(Y)$.

Claim: $H(X) = H(X, f(X)) \geq H(f(X))$ with equality iff f is injective on

supp X . Proof: $H(X) = H(X, f(X)) \geq H(f(X))$

$$\begin{aligned}
H(X, f(X)) &= \sum_x \sum_y P_{XY}(x, y) \log \frac{1}{P_{XY}(x, y)} \\
&= \sum_x \sum_y P_{XX}(x, f^{-1}(y)) \log \frac{1}{P_{XX}(x, f^{-1}(y))} \\
&= \sum_x \sum_y P_X(x) \mathbf{1}(x \in f^{-1}(y)) \log \frac{1}{P_X(x) \mathbf{1}(x \in f^{-1}(y))} \\
&= \sum_x \sum_{y=f(x)} P_X(x) \mathbf{1}(x \in f^{-1}(y)) \log \frac{1}{P_X(x) \mathbf{1}(x \in f^{-1}(y))} \\
&= \sum_x P_X(x) \log \frac{1}{P_X(x)} = H(X).
\end{aligned}$$

Lemma: Let $x_1, x_2, \dots, x_n > 0$ for $n \geq 2$. Then

$$\sum_i x_i \cdot \log \frac{1}{\sum_i x_i} < \sum_i x_i \log \frac{1}{x_i}.$$

Proof: Write equivalently,

$$\sum_i x_i \log x_i < \sum_i x_i \cdot \log \left(\sum_i x_i \right).$$

Then (if f is not one-to-one on supp X):

$$\begin{aligned}
H(f(X)) &= \sum_{y \in f(\text{supp } X)} P_X(f^{-1}(y)) \log \frac{1}{P_X(f^{-1}(y))} \\
&< \sum_{y \in f(\text{supp } X)} \sum_{x \in f^{-1}(y)} P_X(x) \log \frac{1}{P_X(x)} \\
&= \sum_{x \in X} P_X(x) \log \frac{1}{P_X(x)} = H(X).
\end{aligned}$$

$$\begin{aligned}
H(X_1, X_2, X_3) &= H(X_1, X_2) + H(X_3 | X_1, X_2) \\
&= H(X_1) + H(X_2 | X_1) + H(X_3 | X_1, X_2).
\end{aligned}$$

2 Cross-Entropy

Let $p, q : X \rightarrow I$ be distributions. Define the cross-entropy $H(p, q) = -\sum_x p(x) \log_2 q(x)$. Note that $-\log y$ is convex, so

$$H(p, q) \geq -\log_2 \left(\sum_x p(x) q(x) \right).$$

We have

$$\begin{aligned} H(p, q) - H(p) &= - \sum_x p(x) \log_2 q(x) + \sum_x p(x) \log_2 p(x) \\ &= - \sum_x p(x) \log_2 (q(x)/p(x)) \geq - \log_2 \left(\sum_x q(x) \right) = 0, \end{aligned}$$

since $-\log x$ is convex.

Equality holds iff $q(x)/p(x)$ is constant over $x \in X$, ie $p(x) = q(x)$ for all $x \in X$.